

Revised Generation Interconnection System Impact Study Report

For Queue Project Q11-0065

Blount Street 138 kV
October 03, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0065. The Interconnection Customer proposes a 59.2 MW solar photovoltaic facility interconnecting at the Blount Street 138 kV substation in the Pennsylvania area. The request was received on August 21, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 49 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,150,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Wallenpaupack Substation 500/345kV Transformer Bank is overloaded to 101.4% of its 910 MVA emergency rating for the loss of the Tombstone to Bixby Hill Substation 500 KV. The project contributes 8.0% DFAX (4.8 MW) to this overload.

The Sorrell Hill to Pearl 138 KV is overloaded to 101.2% of its 184 MVA emergency rating for the loss of the Peckville to Pearl 138 KV. The project contributes 13.8% DFAX (8.1 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Wallenpaupack Substation 500/345kV Transformer Bank	Replace transformer with higher-rated unit	\$30,620,000	6% (\$1,869,000)
Sorrell Hill to Pearl 138 KV Line	Reconductor line	\$1,281,000	100% (\$1,281,000)
	Total allocated to project		\$3,150,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.